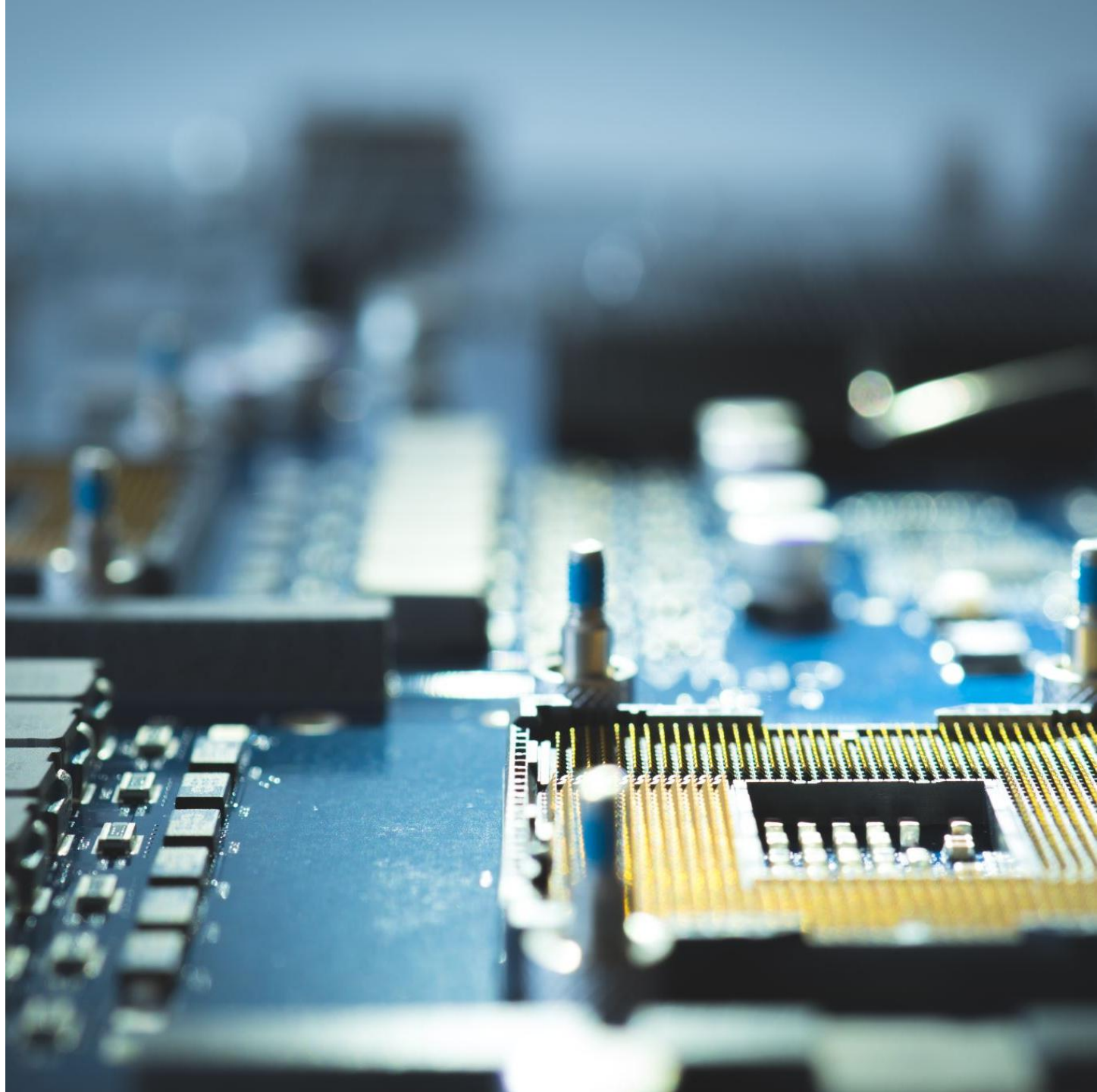




WorkshopPLUS: SharePoint Server Administration

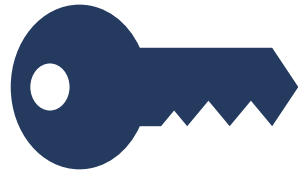
Planning patch management



Agenda - Chapter 1

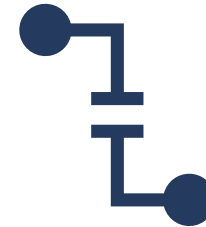
1	Patching Prerequisites
2	Understand the Various Types of Patches
3	Patching in SharePoint
4	Patching Improvements

Required Permissions



Required permissions

securityadmin fixed SQL server role
db_owner fixed database role on all
SharePoint databases
Local administrator on the SharePoint
servers



Important conditions

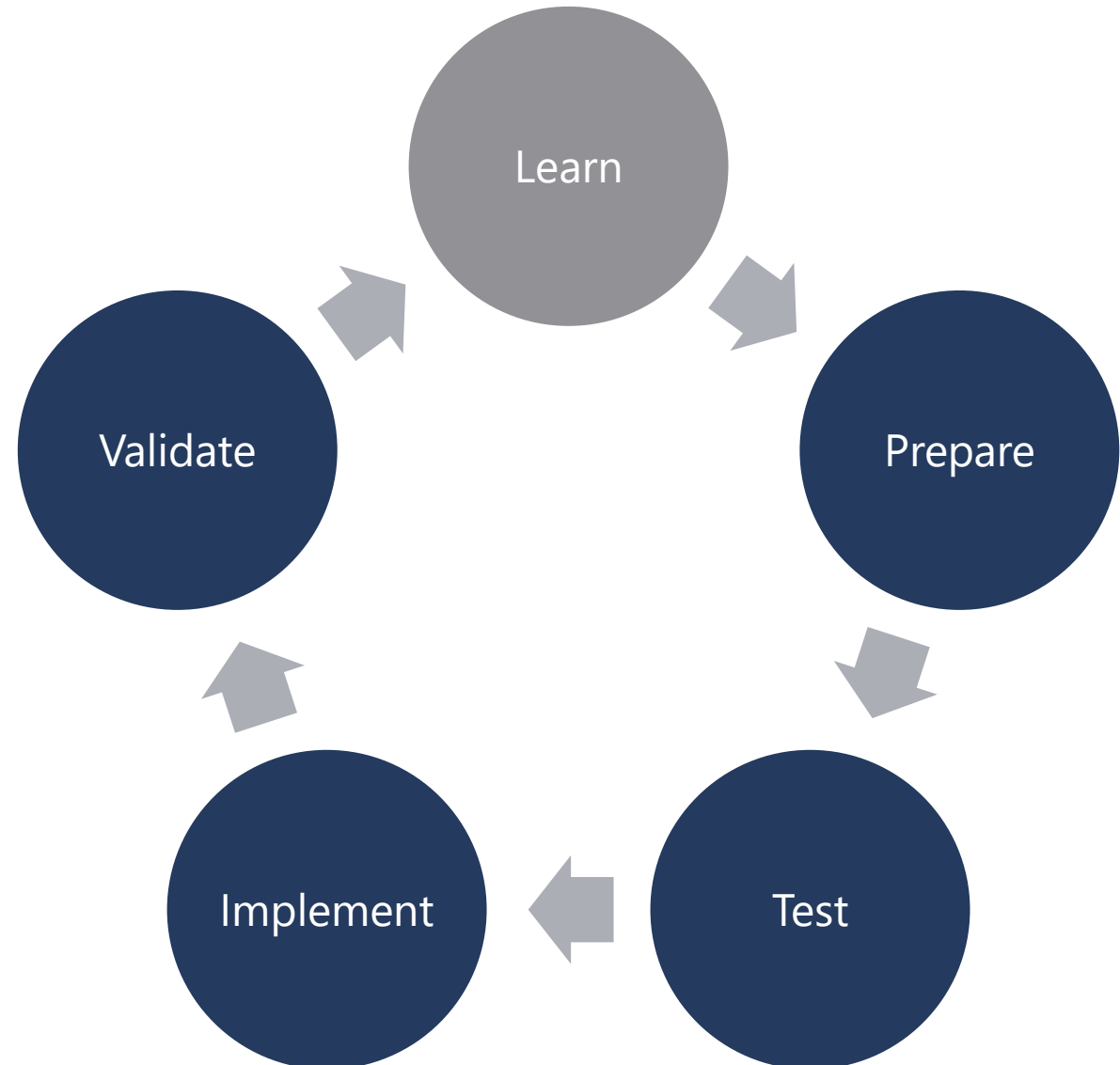
All web servers are load balanced and in
rotation with the load balancer.
All farm servers are operating properly.
All databases are active and operating
properly.

Software Update Deployment Cycle* - Learn

Learn

- Requirements and prerequisites
- Update strategy
- Downtime reduction
- Common issues

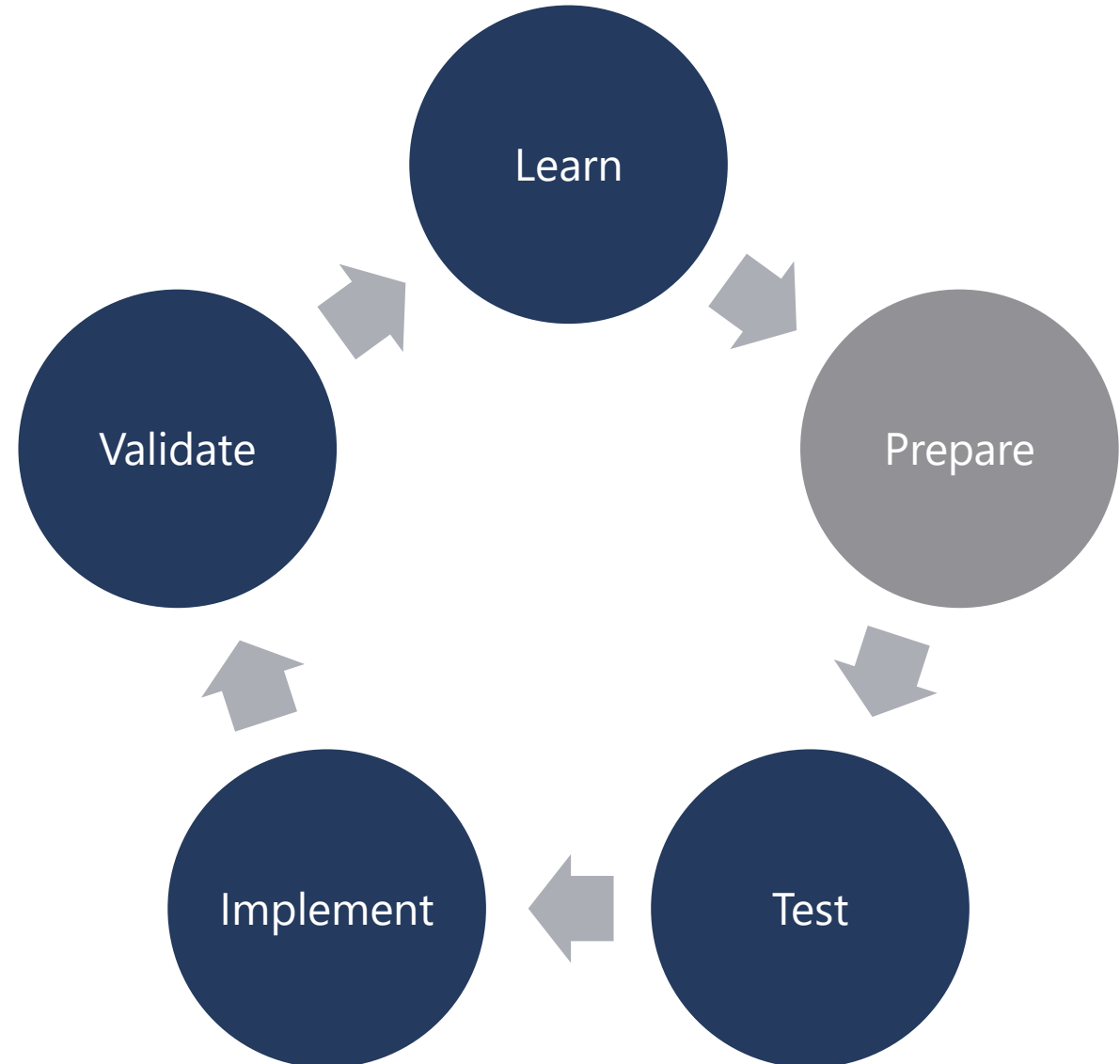
*Applicable to any software installation.



Software Update Deployment Cycle - Prepare

Prepare

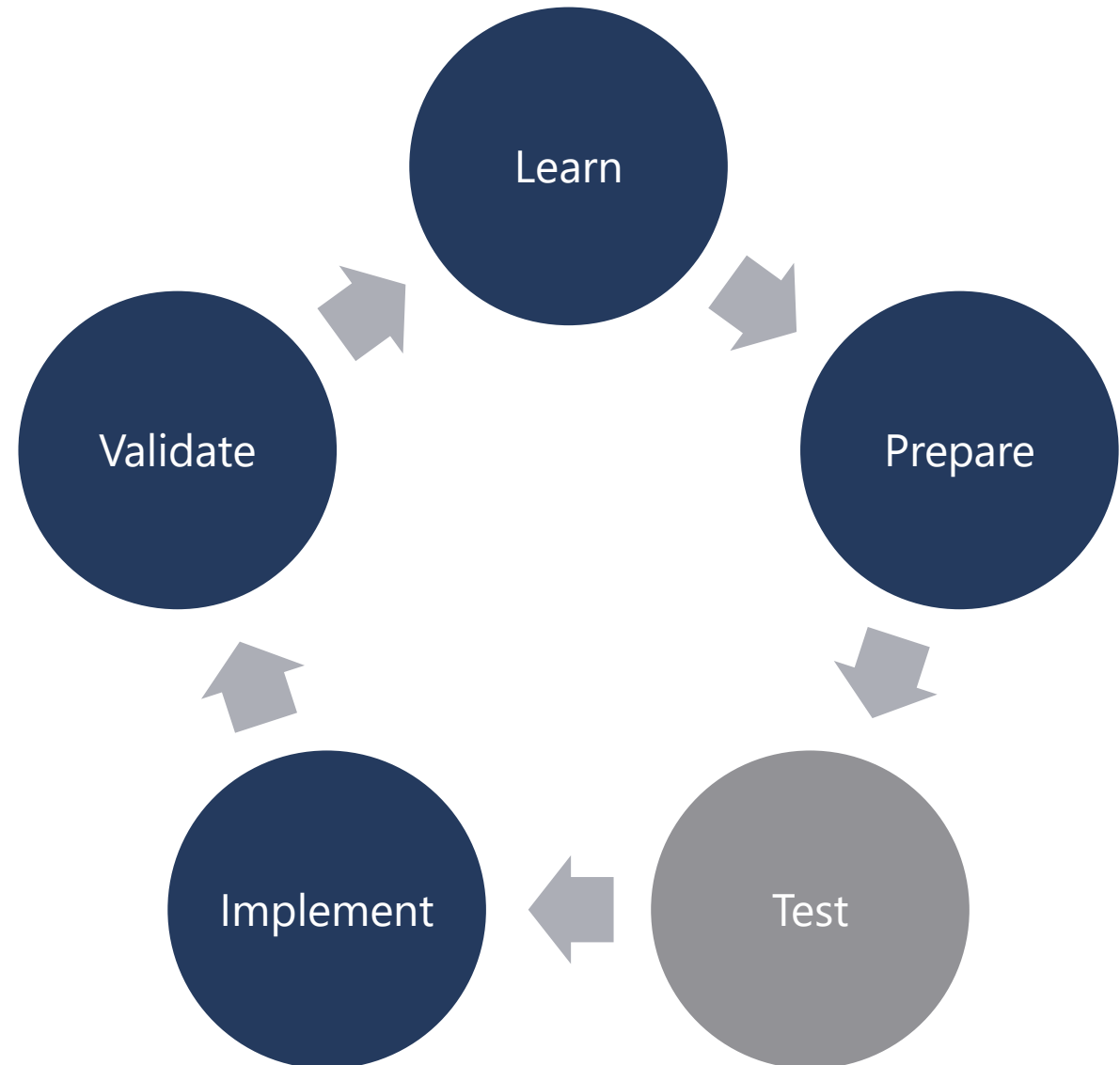
- Document the environment
- Manage Customizations
- Plan the Upgrade Strategy
- Make farm items ready for updates



Software Update Deployment Cycle - Test

Test

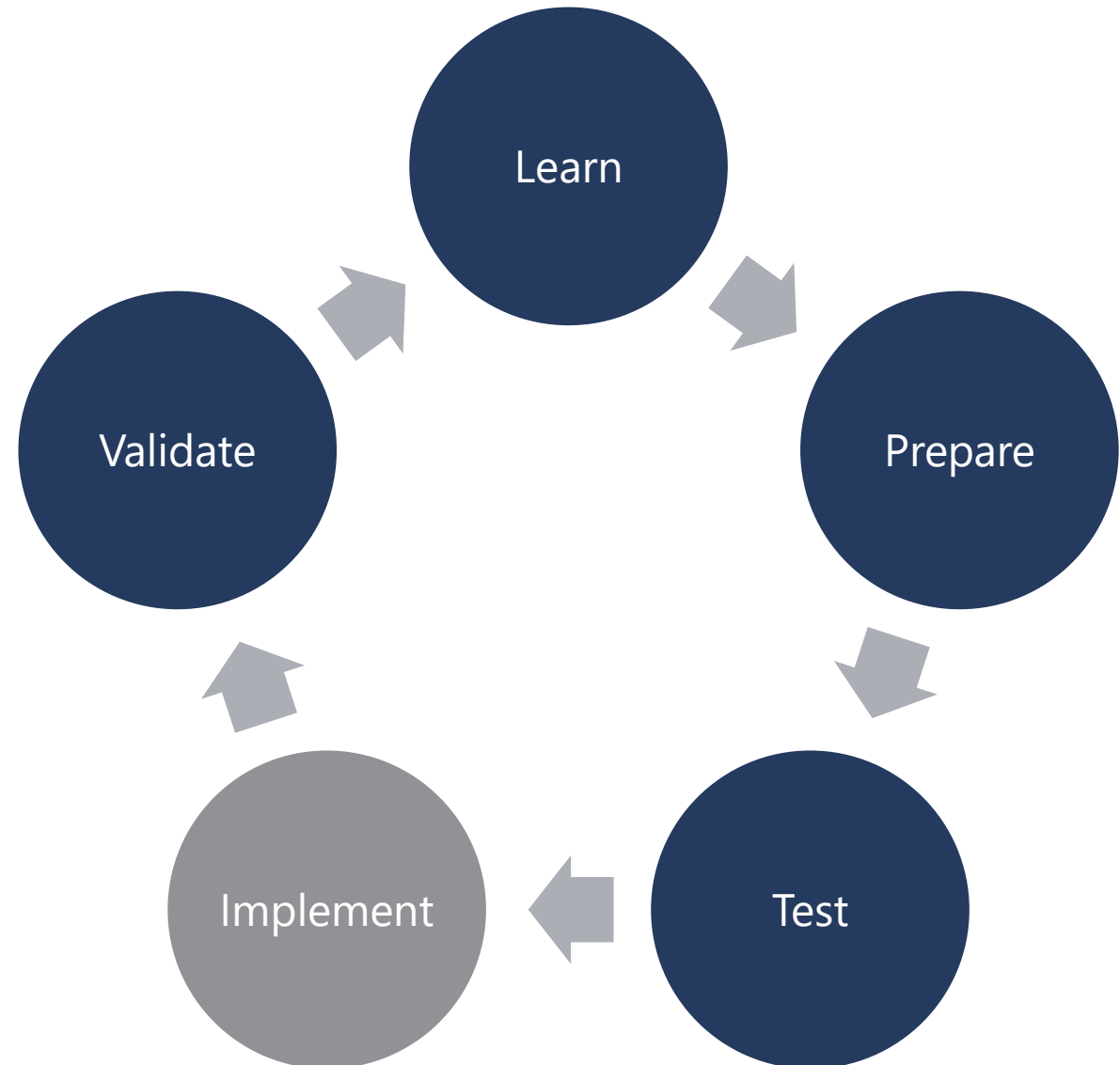
- Build a test farm
- Evaluate techniques



Software Update Deployment Cycle - Implement

Implement

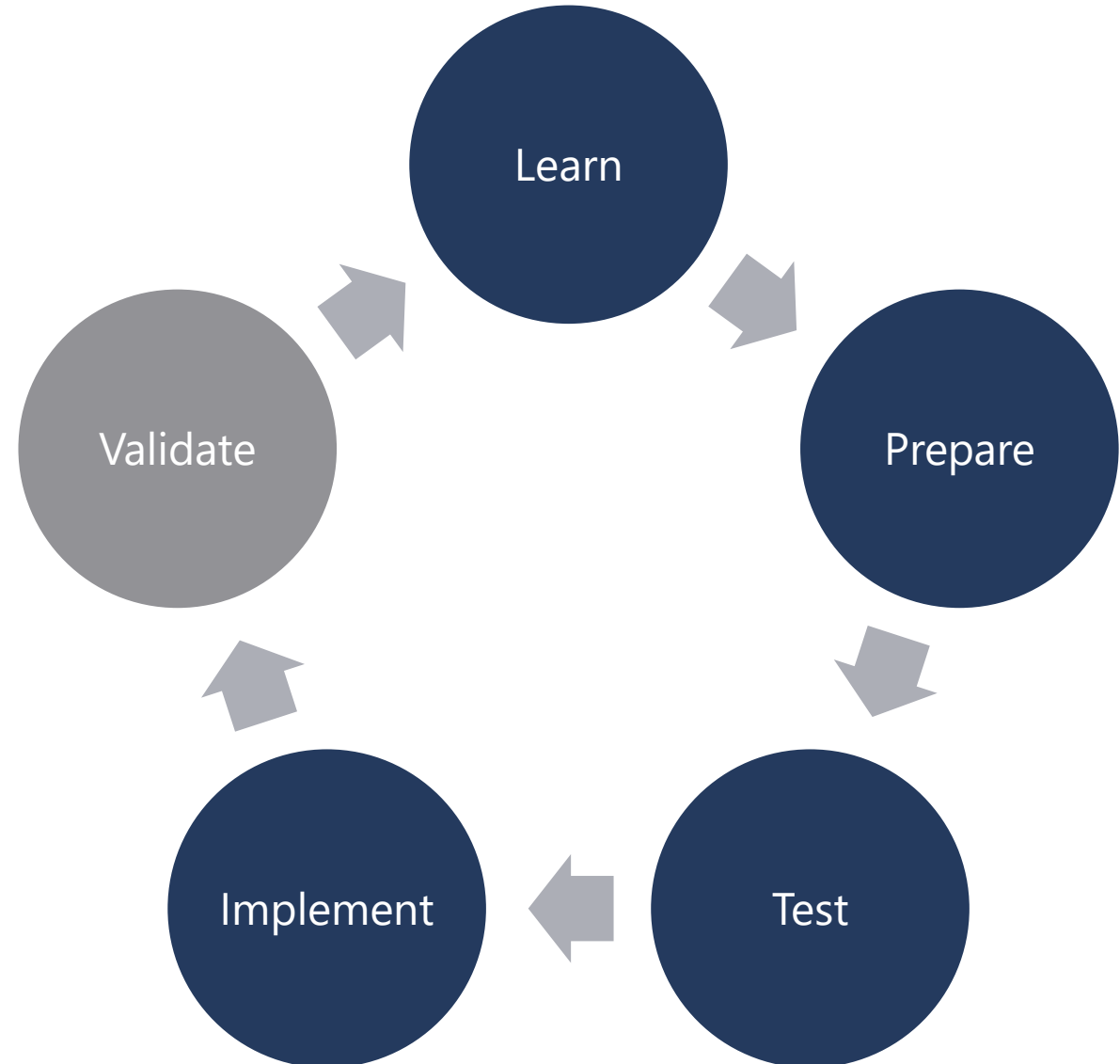
- Build or update farms
- Deploy customizations
- Reduce downtime
- Monitor progress



Software Update Deployment Cycle - Validate

Validate

- Logged event failures
- User interface or experience issues
- Data issues



Agenda - Chapter 2

1

Patching Prerequisites

2

Understand the Various Types of Patches

3

Patching in SharePoint

4

Patching Improvements

Understanding Types of Patches - Updates



Security Updates/Public Updates



Cumulative Updates



Server/User Updates

Understanding Types of Patches - Packs



Service Packs

A tested, cumulative set of all hotfixes, security updates, critical updates, and updates. Additionally, service packs may contain additional fixes for problems that are found internally since the release of the product.



Language Packs

Language packs enable site owners and site collection administrators to create SharePoint sites and site collections in multiple languages without requiring separate installations of SharePoint.



Feature Packs

Feature Packs allow Microsoft to accelerate delivery of cloud-first features to our Software Assurance customers (where noted) of SharePoint Server outside of the traditional 2- to 3-year release cadence.

Agenda - Chapter 3

1

Patching Prerequisites

2

Understand the Various Types of Patches

3

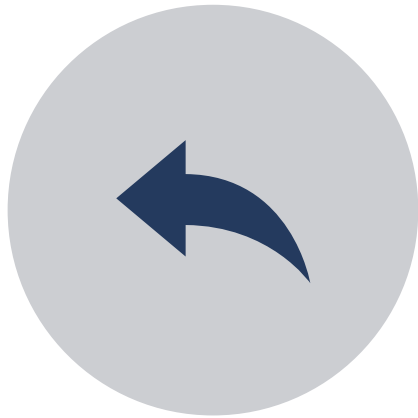
Patching in SharePoint

4

Patching Improvements

Review of Patching In Previous SharePoint Versions

SharePoint Server 2010 vs SharePoint Server 2013



**Support for backward
compatibility**



**Run PSCONFIG on
multiple servers**



Monitor status

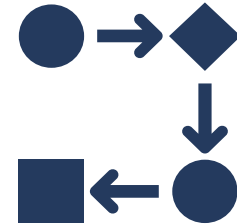
Updating Related Components - AppFabric

AppFabric



SharePoint Only

SharePoint Updates do not update
AppFabric or other components



Procedure

Shut down the service instance (Using
PowerShell command)

Patch AppFabric 1.1 (with Cumulative
Update)

Post-patch operations (may changes be
applied on configuration files)

Start the service instance

Updating Related Components - Workflow

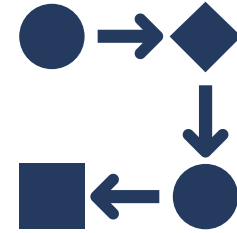
Workflow



Workflow Manager

Workflow is handled by **Workflow Manager 1.0** which is dependent on **Service Bus 1.0**

You need to install Cumulative Updates for each product.



Procedure

Ensure you are logged in as the same admin account you used to install Workflow Manager 1.0.

Run PowerShell command **Register-SPWorkflowService -Force** to pair again with SharePoint.

Updating Related Components - Operating System

Operating System



**Service Packs
and Security
Updates**



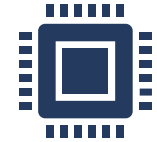
**.NET
Framework**



**Internet
Information
Server (IIS)**



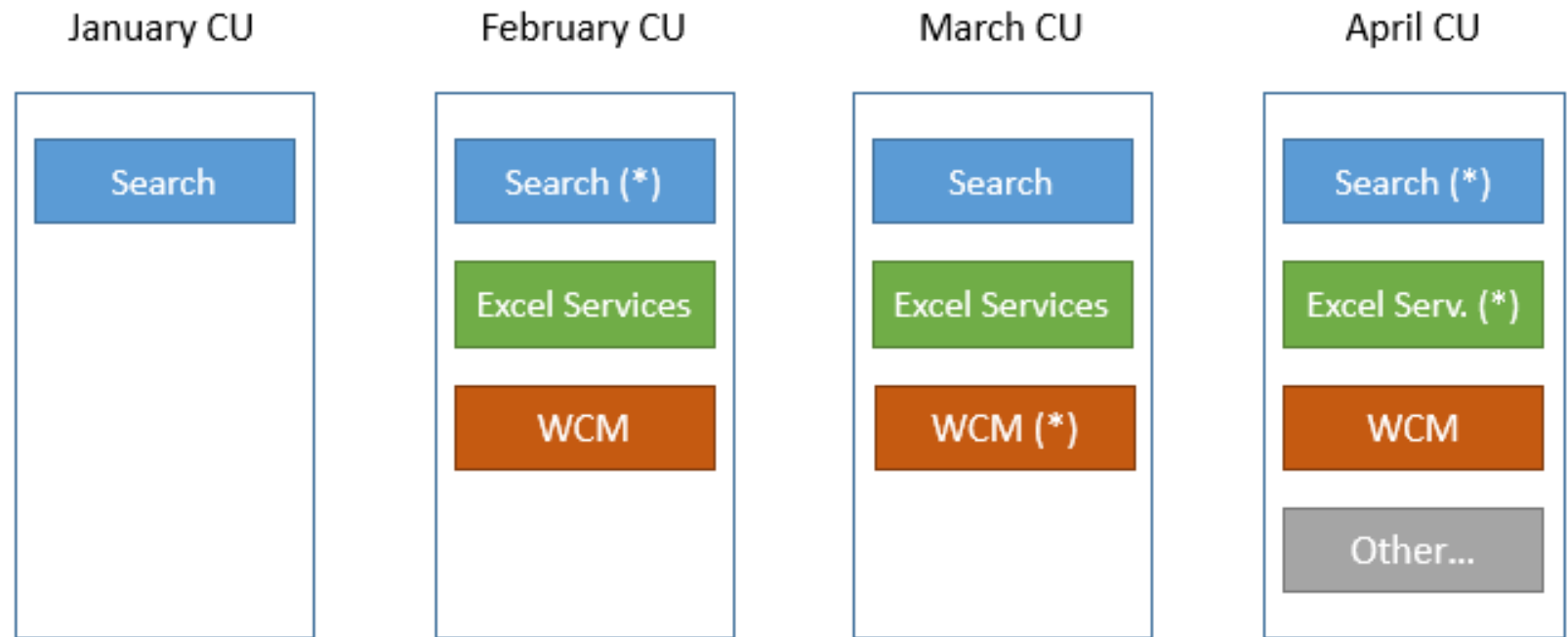
Drivers



Firmware

Role of Uber Packages

- Also called a "full server" package
- Kind of a "mini-service pack"



Simplified Uber Package Model
(*) = fixes released in an earlier CU

Agenda - Chapter 4

1

Patching Prerequisites

2

Understand the Various Types of Patches

3

Patching in SharePoint

4

Patching Improvements

Zero-Downtime Patching



More than one server per role to guarantee service availability

SharePoint Server MinRole can help with this.



Windows Services still have to be restarted



Databases are fully accessible during patching



Increased SQL utilization might affect performance

MSIs and MSPs



Reduction in number of MSIs and MSPs for each patch



Patching Time will be much shorter

Knowledge Check

What are the five steps of a software deployment process?

- Learn, Prepare, Test, Implement, Validate

Give some examples of types of patches:

- Security Updates, Cumulative Updates, Uber Updates, Feature Updates

Questions?



